Load-bearing angle bracket

Request: a bracket that holds 2 kN

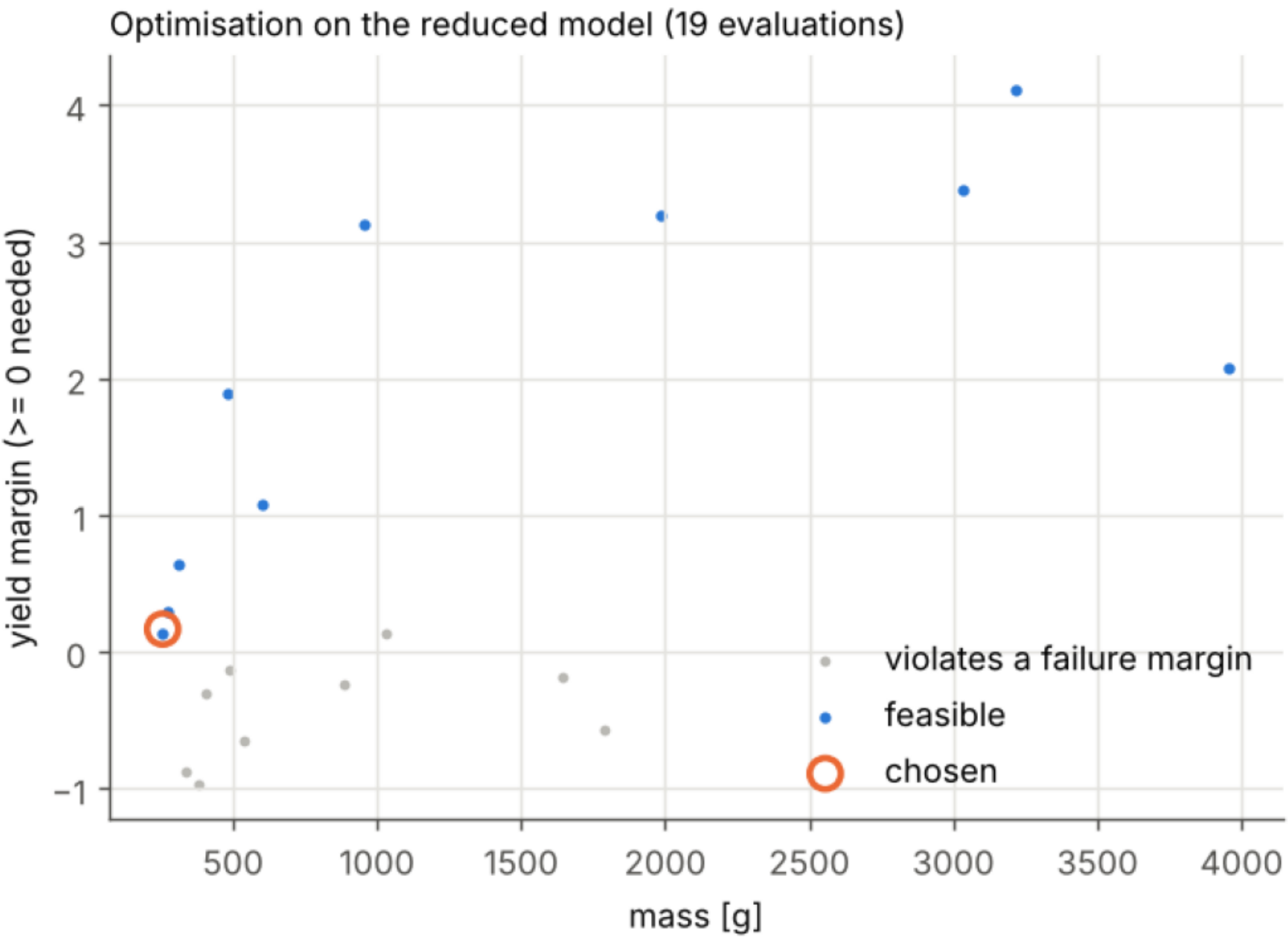
Verdict: will work   P = 0.999 (95 % CI 0.996-0.999)   trust: APPROVED

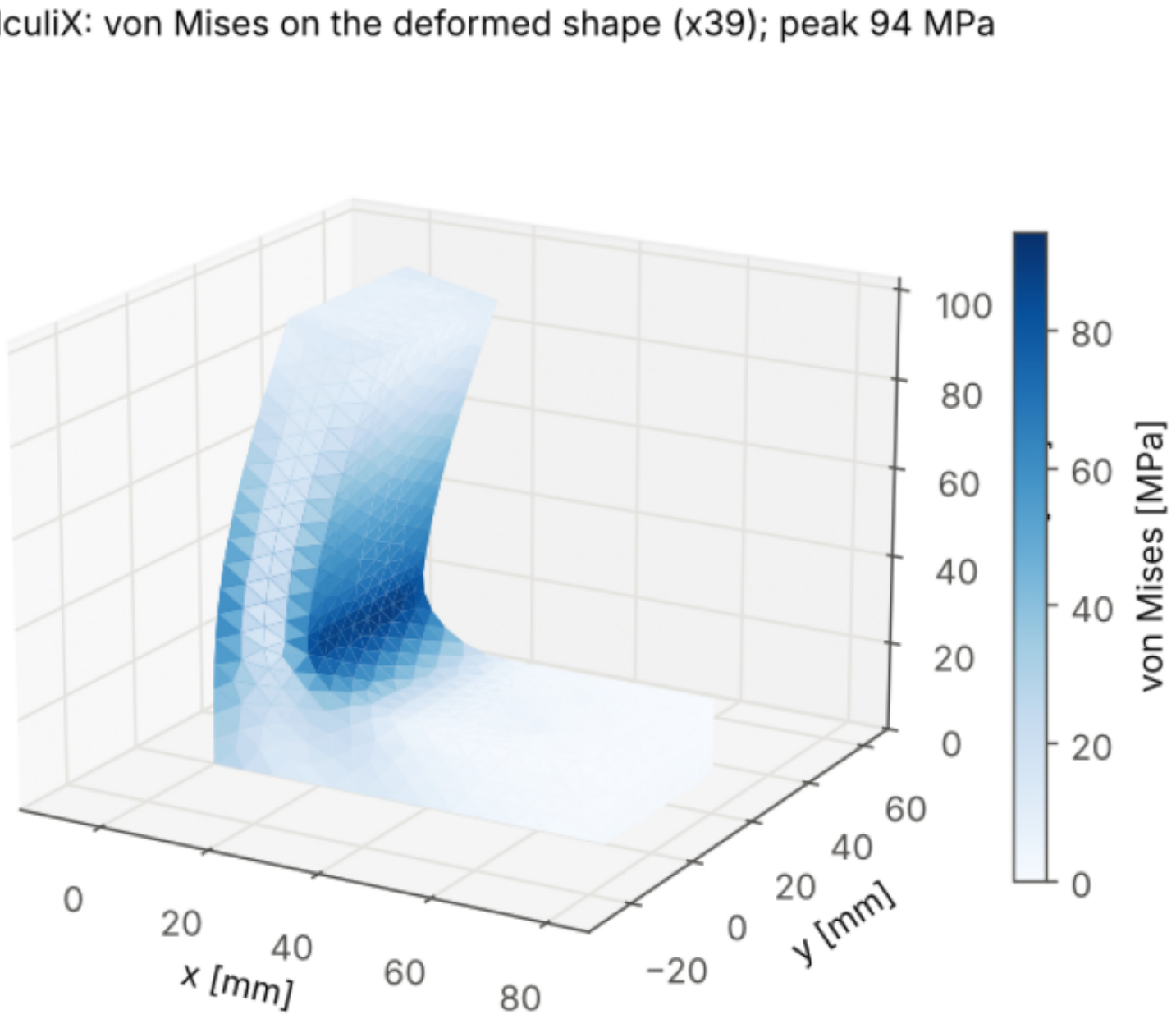
Design:
   plate thickness t: 17 mm
   width w: 36 mm
   mounting-leg length a: 72 mm
   material: al6061_t6

Numbers (fidelity):
   bolts stay below their proof load: 0.01743   [hexahedral FE (SolidFEM)]
   deflection_mm: 0.4383 mm [hexahedral FE (SolidFEM)]
   sigma_peak_mpa: 106.4 MPa [hexahedral FE (SolidFEM)]
   sigma_nominal_mpa: 109.2 MPa [hexahedral FE (SolidFEM)]
   deflection within the stated limit: 1   [hexahedral FE (SolidFEM)]
   bolt_tension_n: 5729 N [hexahedral FE (SolidFEM)]
   no yield at the most stressed point: 0.1673   [hexahedral FE (SolidFEM)]
   mass_g: 256.2 g [hexahedral FE (SolidFEM)]
   no yield at the most stressed point: 0.1491   [cantilever beam theory]
   bolts stay below their proof load: 0.01743   [cantilever beam theory]
   deflection within the stated limit: 1   [cantilever beam theory]
   sigma_peak_mpa: 94.26 MPa [CalculiX (gmsh C3D10)]
   deflection_mm: 0.3796 mm [CalculiX (gmsh C3D10)]
   margin_yield: 0.464   [CalculiX (gmsh C3D10)]
   margin_bolts: 0.1075   [CalculiX (gmsh C3D10)]
   elements: 1.858e+04   [CalculiX (gmsh C3D10)]

Verification:
   sigma_peak_mpa: full 94.26, reduced 96.76, rel. err 0.02653
   deflection_mm: full 0.3796, reduced 0.3804, rel. err 0.002197

Limits:
   - Static load only: no fatigue, impact, vibration or creep.
   - The mounting face is idealised as fully clamped; a flexible wall or anchor changes the stresses and is not
      checked (nor is the anchor's pull-out).
   - Bolt check: tension from the moment about the toe; prying and shear are not included.
   - Linear elastic: valid while the peak stays below yield (the safety factor keeps it there).


Optimisation on the reduced model (19 evaluations)
yield margin (>= 0 needed)
4
3
2
1
0
−1
500
1000
1500
2000
2500
3000
3500
4000
mass [g]
violates a failure margin
feasible
chosen



CalculiX nodal von Mises (extrapolated and averaged at the nodes); each surface triangle shows the mean of its three corners, so the brightest face sits a little below the peak node value in the title. Peak taken away from the clamped and the loaded faces.

